

Homework (Carolina Reaper)

- Q1.** A musician is mixing two songs with different rhythms. Song A has a rhythm pattern that repeats every 3 minutes, while Song B has a rhythm pattern that repeats every 5 minutes. If the musician wants to create a mix that repeats every x minutes, what system of equations represents this situation?
- (A) $3x + 5x = 15$
(B) $3x - 5x = 0$
(C) $\frac{1}{3}x + \frac{1}{5}x = 1$
(D) $\frac{1}{3}x - \frac{1}{5}x = 0$
(E) $3x \cdot 5x = 15$
- Q2.** A painter is creating a mural on a canvas that is 12 feet long and 8 feet wide. If the painter wants to divide the canvas into sections with equal areas, and each section is x feet by y feet, what system of equations represents this situation?
- (A) $xy = 12$
(B) $xy = 8$
(C) $x + y = 12$
(D) $x + y = 8$
(E) $xy = 96$
- Q3.** A music festival has two stages, A and B. Stage A has a sound system that can produce 120 decibels, while Stage B has a sound system that can produce 150 decibels. If the festival organizers want to create a sound mix that is x decibels on Stage A and y decibels on Stage B, and the total sound level is 240 decibels, what system of equations represents this situation?
- (A) $x + y = 240$
(B) $x - y = 30$
(C) $x + y = 270$
(D) $x - y = 60$
(E) $x + y = 210$
- Q4.** A sculptor is creating a sculpture with two types of materials, A and B. Material A costs 15 per pound, while Material B costs 20 per pound. If the sculptor wants to create a sculpture that weighs x pounds of Material A and y pounds of Material B, and the total cost is 300, what system of equations represents this situation?
- (A) $15x + 20y = 300$
(B) $15x - 20y = 0$
(C) $15x + 20y = 250$
(D) $15x - 20y = 50$
(E) $15x + 20y = 350$
- Q5.** A photographer is taking pictures of a cityscape with two cameras, A and B. Camera A has a lens that can capture 30
(A) $0.3x + 0.4y = 0.6$
(B) $0.3x - 0.4y = 0$
(C) $0.3x + 0.4y = 0.7$
(D) $0.3x - 0.4y = 0.1$
(E) $0.3x + 0.4y = 0.5$
- Q6.** A composer is creating a song with two melodies, A and B. Melody A has a frequency of 200 Hz, while Melody B has a frequency of 250 Hz. If the composer wants to create a mix that has a frequency of x Hz for Melody A and y Hz for Melody B, and the total frequency is 420 Hz, what system of equations represents this situation?
- (A) $x + y = 420$
(B) $x - y = 50$
(C) $x + y = 450$
(D) $x - y = 100$
(E) $x + y = 400$
- Q7.** A dancer is creating a routine with two types of steps, A and B. Step A takes 2 seconds to complete, while Step B takes 3 seconds to complete. If the dancer wants to create a routine that takes x seconds to complete with Step A and y seconds to complete with Step B, and the total time is 30 seconds, what system of equations represents this situation?
- (A) $2x + 3y = 30$
(B) $2x - 3y = 0$
(C) $2x + 3y = 25$
(D) $2x - 3y = 5$
(E) $2x + 3y = 35$
- Q8.** A poet is writing a poem with two types of lines, A and B. Line A has 5 syllables, while Line B has 7 syllables. If the poet wants to create a poem with x lines of Type A and y lines of Type B, and the total number of syllables is 60, what system of equations represents this situation?
- (A) $5x + 7y = 60$
(B) $5x - 7y = 0$
(C) $5x + 7y = 55$
(D) $5x - 7y = 10$
(E) $5x + 7y = 65$

Open Ended Questions (FRQ)

- Q9.** A music producer is creating a song with two instruments, A and B. Instrument A plays a note that is 2 octaves higher than Instrument B. If the frequency of Instrument B is 200 Hz, and the total frequency of the song is 500 Hz, find the frequency of Instrument A and the system of equations that represents this situation. Show your work and explain your answer.
- Q10.** A painter is creating a mural with two colors, A and B. Color A takes 3 hours to dry, while Color B takes 2 hours to dry. If the painter wants to create a mural that takes x hours to dry with Color A and y hours to dry with Color B, and the total time to dry is 10 hours, find the system of equations that represents this situation and the values of x and y . Show your work and explain your answer.

Chef's Tips

- Tip 1: Make sure to read the questions carefully and understand the context.
- Tip 2: Use the system of equations to represent the situation, and then solve for the unknowns.
- Tip 3: Check your work and explain your answer clearly.